

Mini horn telescope construction and operation

Construction and observing guide

Build two mini horn telescopes, connect their receivers, and collect hydrogen spectra. Assemble the horn, can receiver, and wooden stand separately before joining them.

Construction materials for two telescopes

Quantity	Material
1	4 × 8-foot sheet of 1/2-inch foil-faced rigid foam insulation
5	2 × 2 boards, each 8 feet long
1 roll each	Aluminum foil tape and masking tape
24	3 1/2-inch bolts with wing nuts
4	3 1/2-inch flat-head wood screws
8	1-inch flat-head wood screws
2	New, empty F-style oblong 1-gallon metal cans
2	4-inch pieces of 4-gauge solid copper grounding wire
2	Four-hole, panel-mount SMA male solder terminals with RF coax adapters
1 roll	Solder

The insulation sheet can provide six horn shells. This parts list covers two complete telescopes.

Scrap materials can reduce costs. Foam pieces must be at least 16 × 48 inches. The original list allows a 6-inch wire offcut for both probes and 40 feet of lumber overall.

Use new, empty metal cans. Do not cut, drill, or heat a container that held flammable liquid.

Tools

Gather a drill and bits, saw, screwdriver, tape measure, box cutter, markers, metal straightedge, and smooth-edge can opener. Add a soldering iron and hot-air soldering tool.

Cut the horn panels

Use the dimensions for this mini horn throughout construction. The separate 2021 horn guides describe a different design.

1. Cut the 4 × 8-foot insulation sheet into two 4 × 4-foot sections. One section supplies enough panels for three horns.
2. Mark four panels using Figure 1. Guide the box cutter with a metal straightedge.
3. Cut the panels carefully, then repeat for the second telescope.

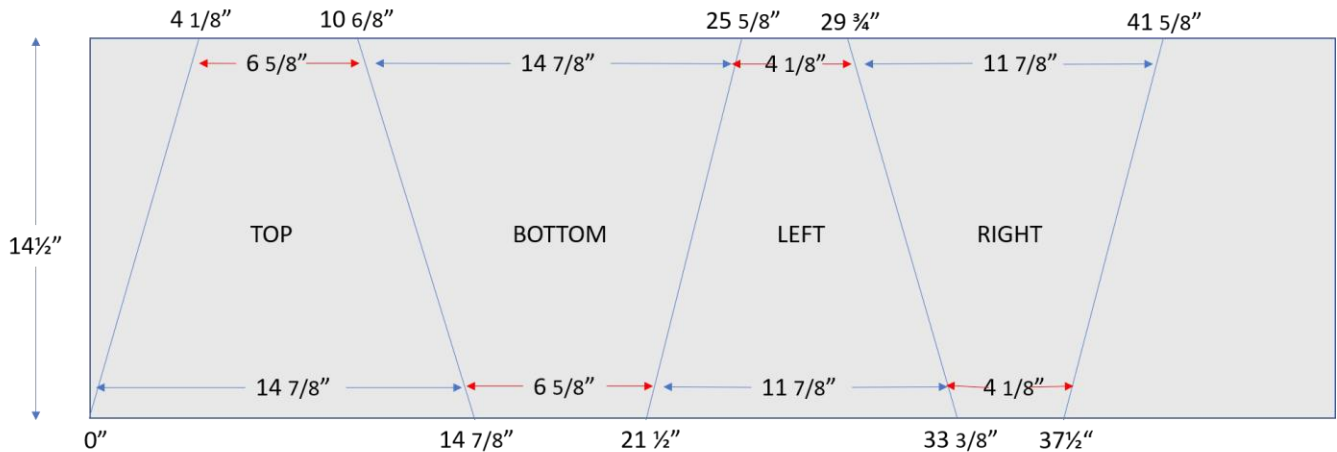


Figure 1. Panel layout and dimensions for the mini horn.

Receiver parts for two telescopes

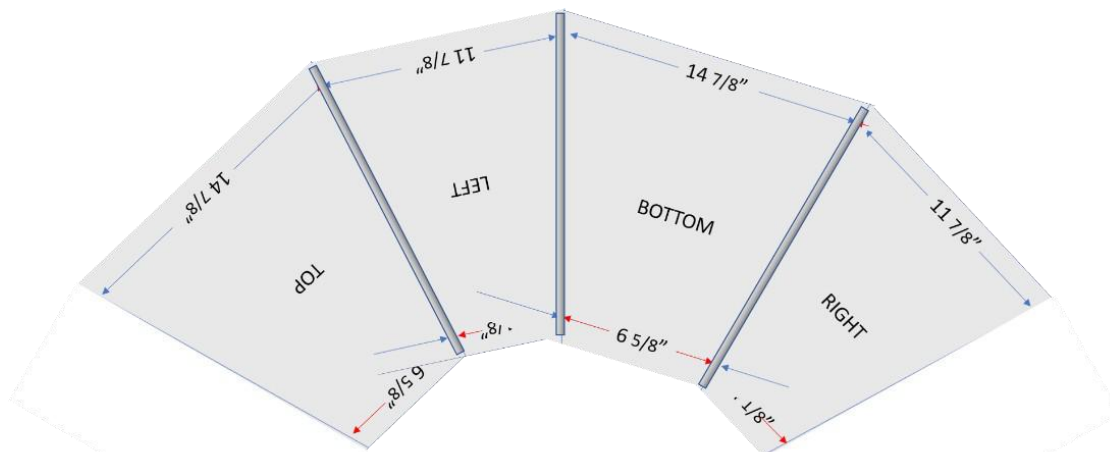
Quantity	Part
2	Airspy Mini software-defined radio receivers
2	Male SMA to male SMA coaxial cables, at least 3 feet long
2	DSPIRA low-noise amplifiers

The low-noise amplifier is called the LNA throughout this guide.

[Request kit equipment from WVU DSPIRA](#)

Tape the horn panels

1. Arrange the four panels as shown below, with foil faces inside the finished horn.
2. Join the three adjoining seams with 2-inch foil tape. Fold the panels until the remaining edges meet.
3. Match inside edge to inside edge. Do not overlap one panel outside the other.
4. Hold the seam with short pieces of tape, then tape its entire inside edge.
5. Tape all outside seams. Seal the top and bottom edges with foil tape.



After cutting, align the panel parts as shown above. Connect the pieces using the aluminum tape. The tape should run the entire length of the seam. Fold over or cut off any extra at the ends. When you have taped the three sides laying flat then fold the panes so that the edge of top aligns with the edge of right. Tape the entire inside seem where these two panels come together. After the cone seems are completely taped on the inside then tape the entire length of the outside seems with the aluminum tape. Lastly, cover the top and bottom edges of all the panels with the aluminum tape to protect them from damage.

Figure 2. Panel arrangement before folding.

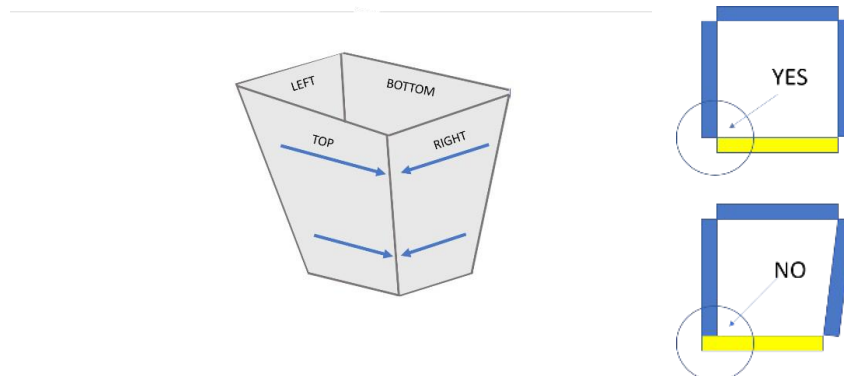


Figure 3. Folded horn and edge alignment. Match the inner edges without overlapping panels.

Prepare the can receiver

1. Remove the can bottom with a smooth-edge can opener. Keep the capped end.
2. Mark a point on the wide side, centered between its edges and 5.25 cm from the capped end.
3. Support the wall with a snug wood block. Drill a 1/8-inch hole, then remove the block.
4. Cover the pour spout with foil tape on both the inside and outside.
5. Position the drilled side upward. Heat the can and apply solder around the hole.
6. Insert the antenna coupler, keeping the copper probe inside and the four-hole plate flat outside.
7. Center the probe so it does not touch the metal can. Heat the solder until it flows beneath the plate.
8. Remove the heat. Let the solder solidify and the can cool before handling it.

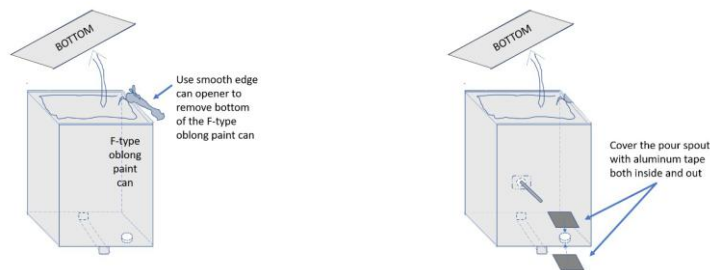


Figure 4. Remove the bottom and prepare the receiver opening.

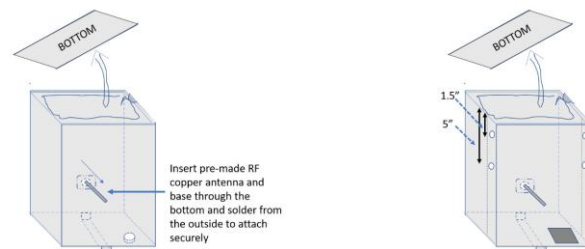


Figure 5. Position and solder the antenna coupler.

Attach the wooden receiver frame

1. Set the can on a table with the antenna connector upward. Place a 2 × 2 board beside it.
2. Use the board as a guide to mark a line on each side of the can.
3. Mark holes 1 1/2 inches and 5 inches from the open end on each side. Center them between the edge and guide line.
4. Support the can with the wood block before drilling the four 1/8-inch holes. Remove the block afterward.
5. Attach the two 20-inch wooden pieces to the can, as shown below.
6. Trim the 8-inch crosspiece to fit between the side pieces, approximately 6 5/8 inches. Attach it to form a U-shaped frame.

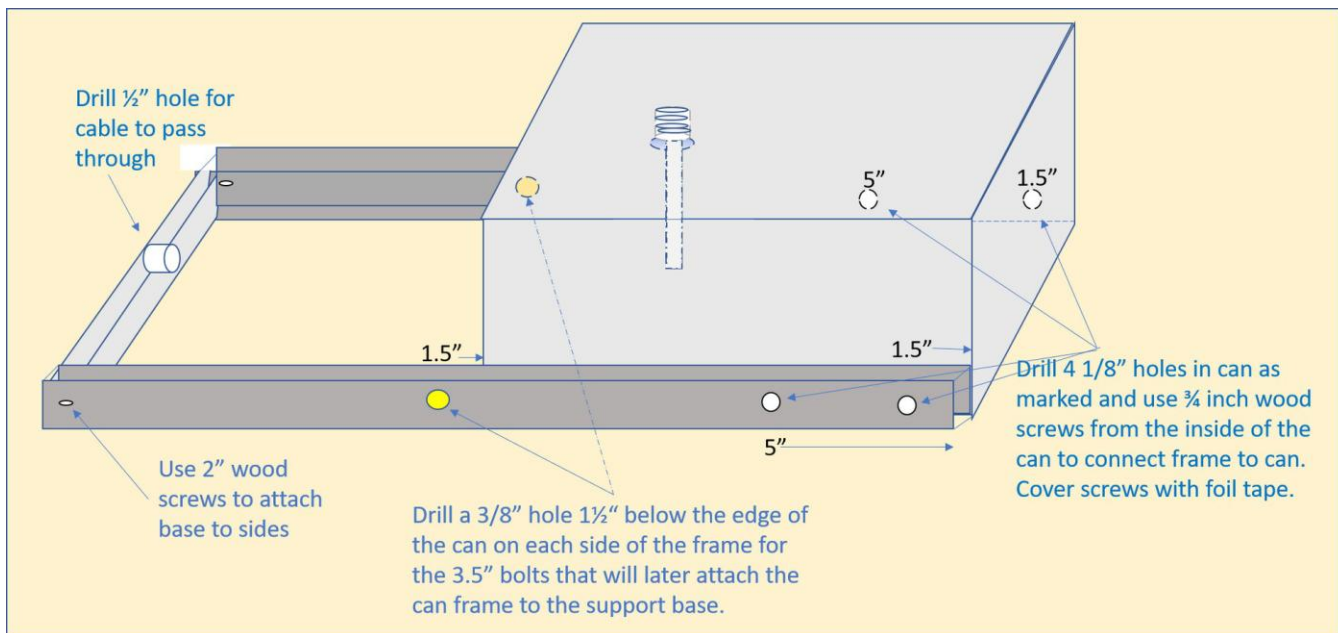


Figure 6. Receiver frame, screw positions, cable opening, and mounting holes.

Follow the diagram for the cable opening and the 3/8-inch stand-mounting holes. Cover screws inside the can with foil tape.

Follow Figure 7 for lengths, overlap cuts, and the receiver and laptop supports. Drill 3/8-inch holes for the 3 1/2-inch bolts.



Figure 8. Locations of the cut pieces in the stand.

Assemble the stand and horn

1. Join the stand pieces at the positions shown in Figure 9. Insert bolts from the outside, with threaded ends inward.
2. Tighten a wing nut on each bolt. Check that the frame stands without twisting or moving.
3. Insert the can through the horn mouth, handle end first. Push it toward the narrow opening until the open rim remains inside.
4. Align a loose opening with the can rim. Hold the joint with short pieces of foil tape.
5. Tape the complete joint inside and outside. Cover the exposed can rim and keep the can from moving.

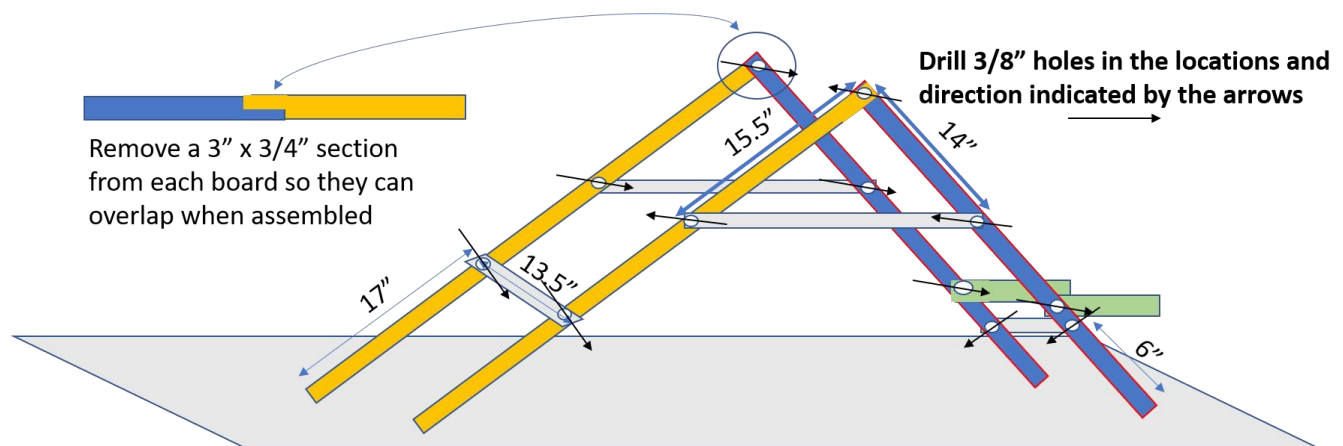


Figure 9. Stand joints and bolt locations.

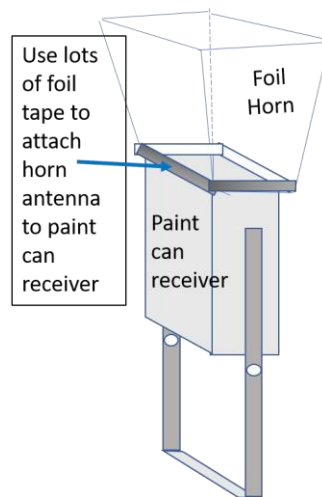


Figure 10. Tape the receiver to the horn around the entire joint.

Mount the horn and connect the receiver

Attach the assembled horn and receiver frame to the stand at the mounting holes shown below.

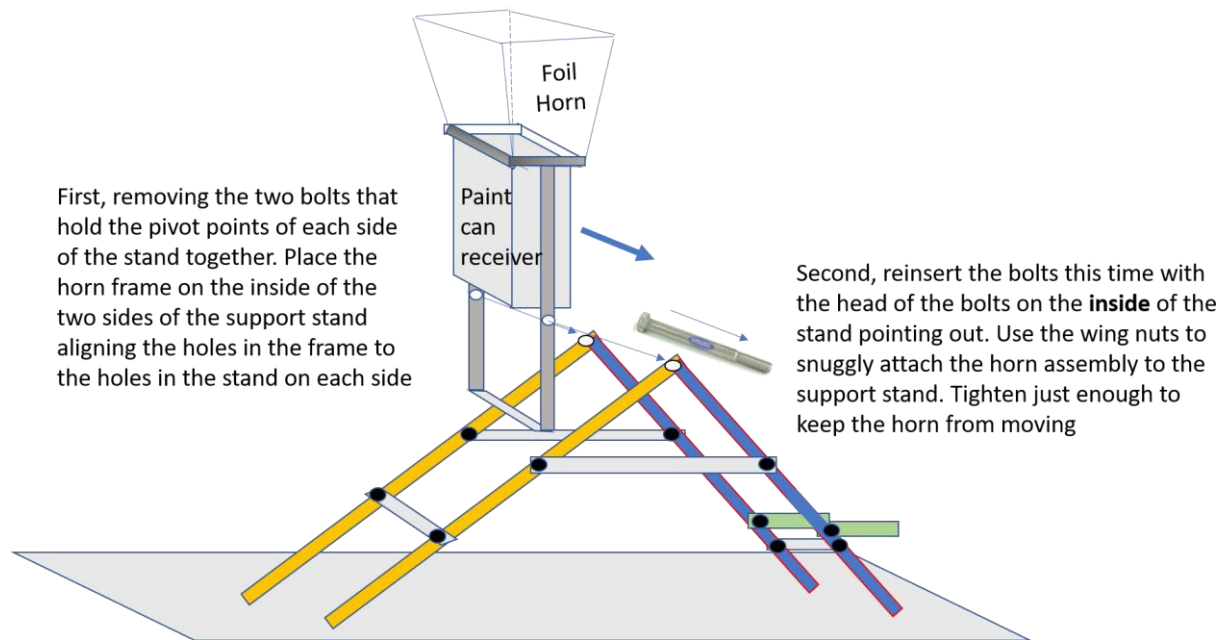


Figure 11. Attach the receiver frame to the stand.

Connect the electronics

1. Attach the LNA input to the SMA connector outside the can. This connector points perpendicular to the amplifier board.
2. Connect the LNA output to the Airspy using the SMA cable. The amplifier output points along the board.
3. Connect the Airspy to the computer with its USB cable. The receiver and computer ends have different connectors.

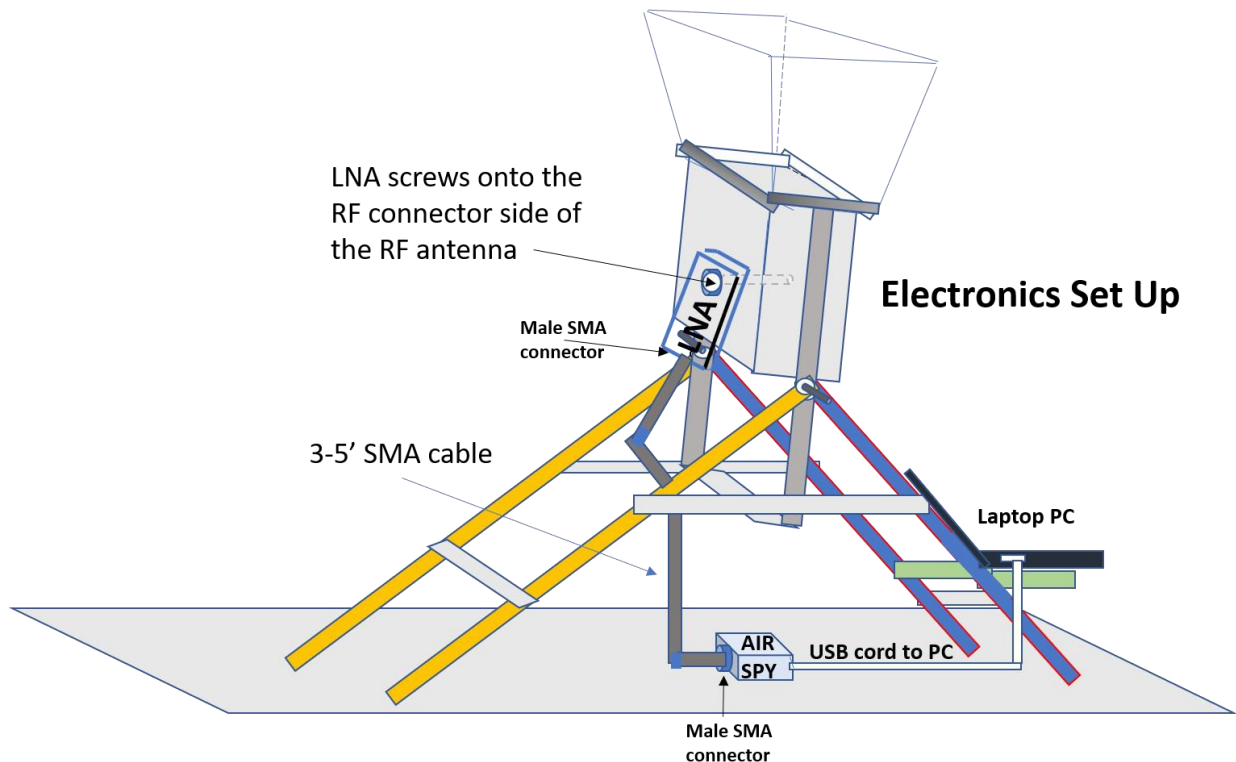


Figure 12. The receiver connections carry the signal from the horn to the computer.

Prepare the observing computer

The current application is calibrated-spectrometer.grc in the DSPIRA software repository. Older images and videos call it spectrometer_w_cal.grc.

[Current software installation guide](#)

[Receiver settings for Airspy and other devices](#)

Use a supplied bootable USB drive

1. Turn off the computer. Connect the receiver and insert the supplied USB drive into a separate port.
2. Start the computer and open its boot menu. F12 works on the computers used in the original guide; other models differ.
3. Select the USB drive. Allow Ubuntu and the desktop to finish loading.
4. Open the spectrometer flowgraph supplied with that image. Allow GNU Radio to load before running it.
5. Check the receiver settings, recording folder, and clock before observing. Older images use grc_data and default to UTC.

Create a replacement USB drive

[DSPIRA bootable images and instructions](#)

1. Choose an image that supports your computer. Check its drive-capacity requirements before downloading.
2. Extract the downloaded ZIP file. Back up the USB drive before writing the image.
3. Use balenaEtcher to write the extracted image. This erases the selected drive.
4. Boot from the USB drive using its instructions. Confirm receiver operation and the recording folder before collecting data.

Screenshots in this guide show the original interface. Use the current setup guides when an installed version differs.

Recognize the hydrogen spectrum

The horn detects radio waves near the neutral hydrogen line at 1420.4 MHz. It also receives local interference and noise from its electronics.

Calibration estimates the system response so the hydrogen feature is easier to measure. It does not remove every source of interference.

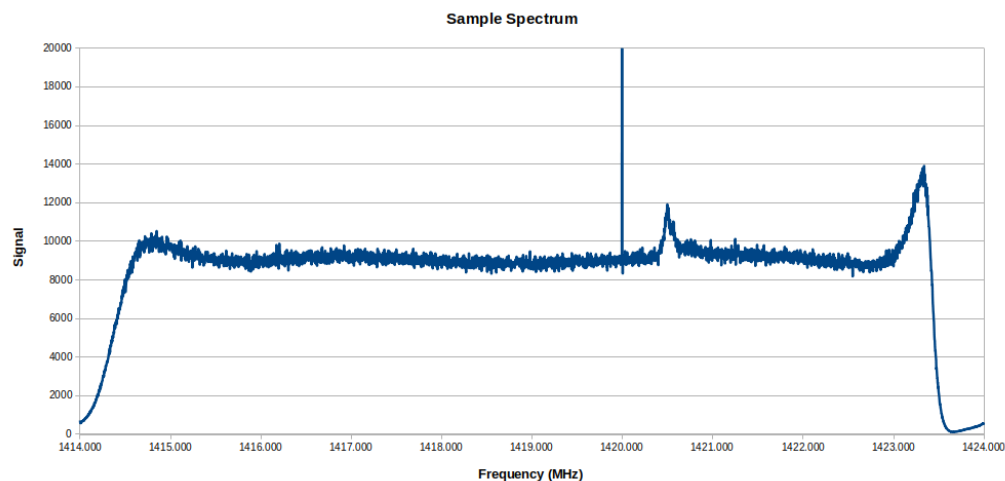


Figure 13. A hydrogen feature appears near 1420.4 MHz in this example spectrum.

The original display spans roughly 1414 to 1424 MHz. The displayed band depends on the receiver and sample-rate settings.

The broad background shape comes partly from the receiver and signal processing. Its level can change as the telescope points at different objects.

An instrumental spike may appear near 1420.0 MHz. Do not confuse that feature with the nearby hydrogen line.

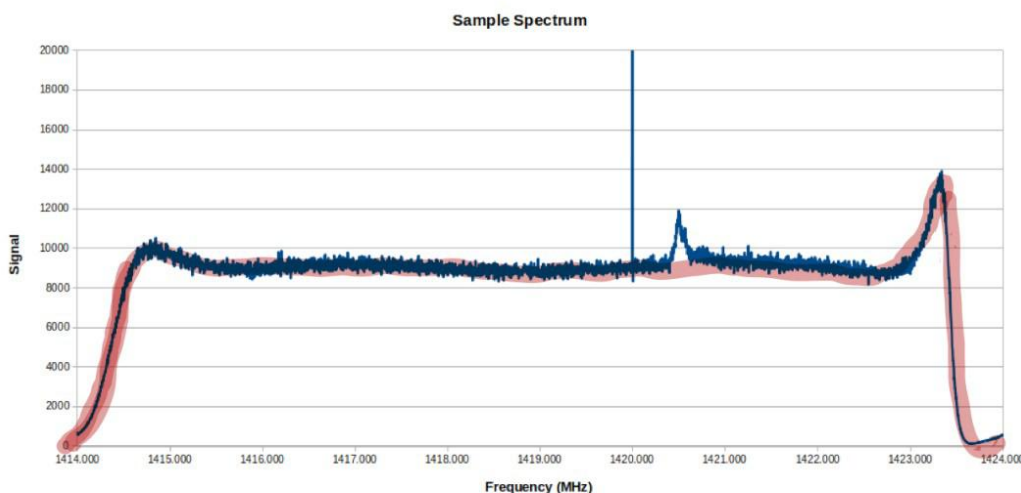


Figure 14. The system background is highlighted in the original example.

Read the Spectrum tab

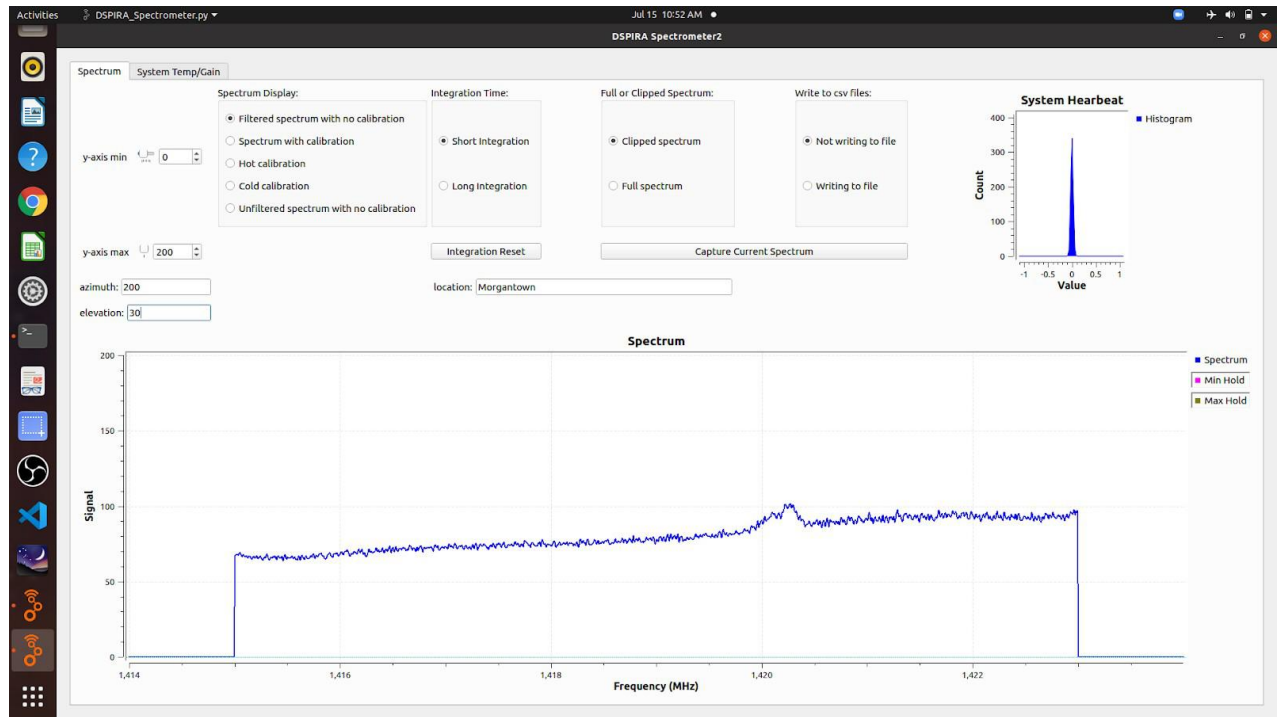


Figure 15. Spectrum display and controls in the original interface.

The main graph plots signal against frequency. The System Heartbeat histogram should keep updating while the program runs.

If the heartbeat freezes, check whether the program has stopped before restarting it.

Display choice	Use
Filtered spectrum with no calibration	View the raw spectrum with noise spikes filtered.
Spectrum with calibration	Apply the saved hot and cold references.
Hot calibration	Collect the ground reference during calibration.
Cold calibration	Collect the sky reference during calibration.
Unfiltered spectrum with no calibration	View the received spectrum without these corrections.

Choose integration and diagnostic displays

Integration averages spectra over time. Longer averages reduce random noise and help reveal weak hydrogen signals.

Control	Behavior
Short Integration	Quick checks and demonstrations; the original default is 0.4 seconds.
Long Integration	Smoother spectra at a settled pointing; the original default is 10 seconds.
Integration Reset	Restart the long average after changing the pointing.
Clipped Spectrum	Hide the distracting band edges. The original display excludes about 1414–1415 and 1423–1424 MHz.
Full Spectrum	Show the full displayed frequency band.

System Temp Gain tab

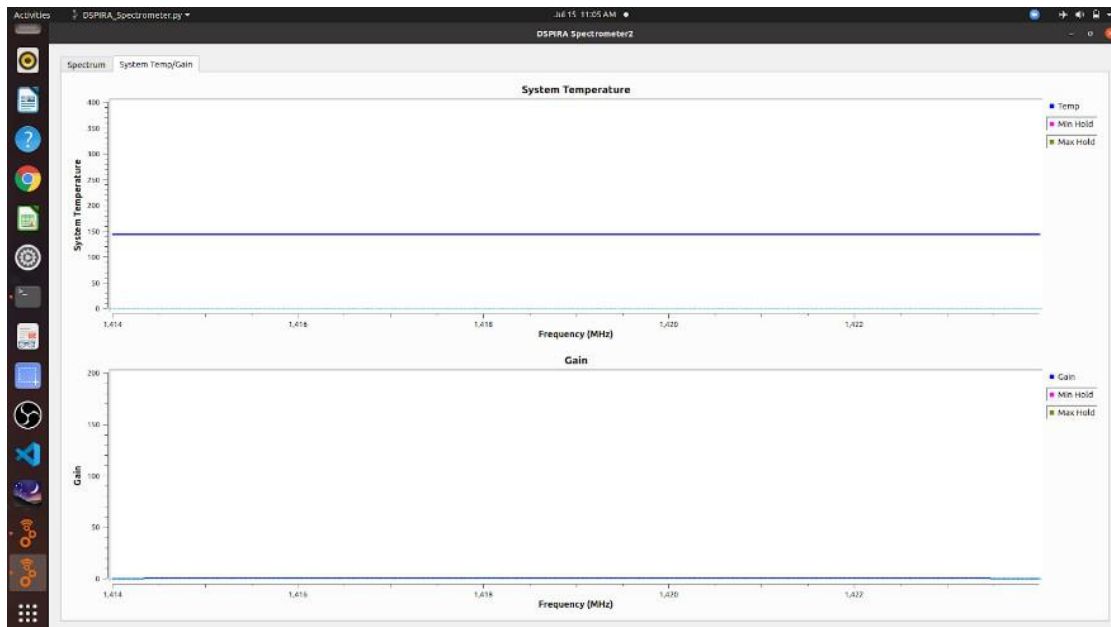


Figure 16. System temperature and gain diagnostic displays.

System Temperature estimates the noise associated with the telescope system. Gain describes the scaling applied to the measured signal.

These diagnostics help assess performance; they are not needed for every observing run.

Set the integration time and output folder

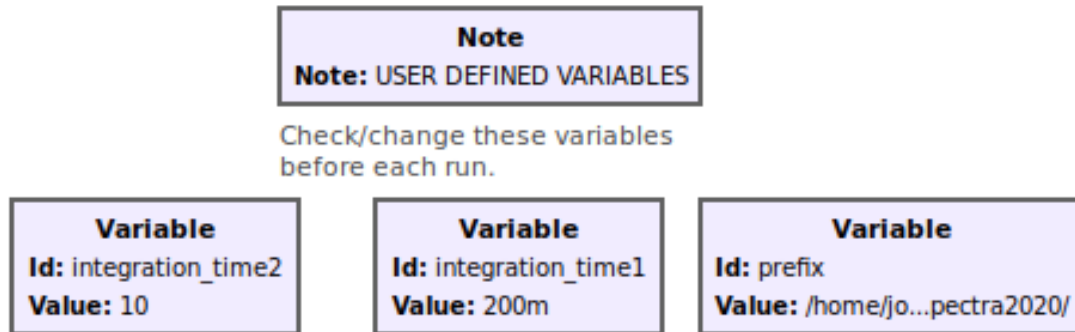


Figure 17. Variable blocks in the original GNU Radio flowgraph.

Before running, open the relevant Variable block by double-clicking it. Change its Value field and save the flowgraph.

- integration_time1 sets the short integration. integration_time2 sets the long integration. Use 0.4 seconds or longer for this flowgraph.
- prefix specifies the output folder. Confirm that it points to a writable location.
- Prevent the computer from sleeping during a long observation. Check its power settings, especially when running on battery.

Choose a recording folder

1. Open the file manager and create or select the folder for your observation.
2. Copy its full path. In older Ubuntu versions, the folder Properties dialog lists the parent path.
3. If you copied only the parent path, append the folder name. Separate directory names with forward slashes.
4. Enter the full path as a quoted string in prefix. End it with a slash.
5. Save a test spectrum and confirm that the file appears in the intended folder.

Example for a folder under the user Jansky:

```
"/home/Jansky/radio_astronomy/horn_data_2020/"
```

Replace the example with an actual folder on your computer.

Start a run and save spectra

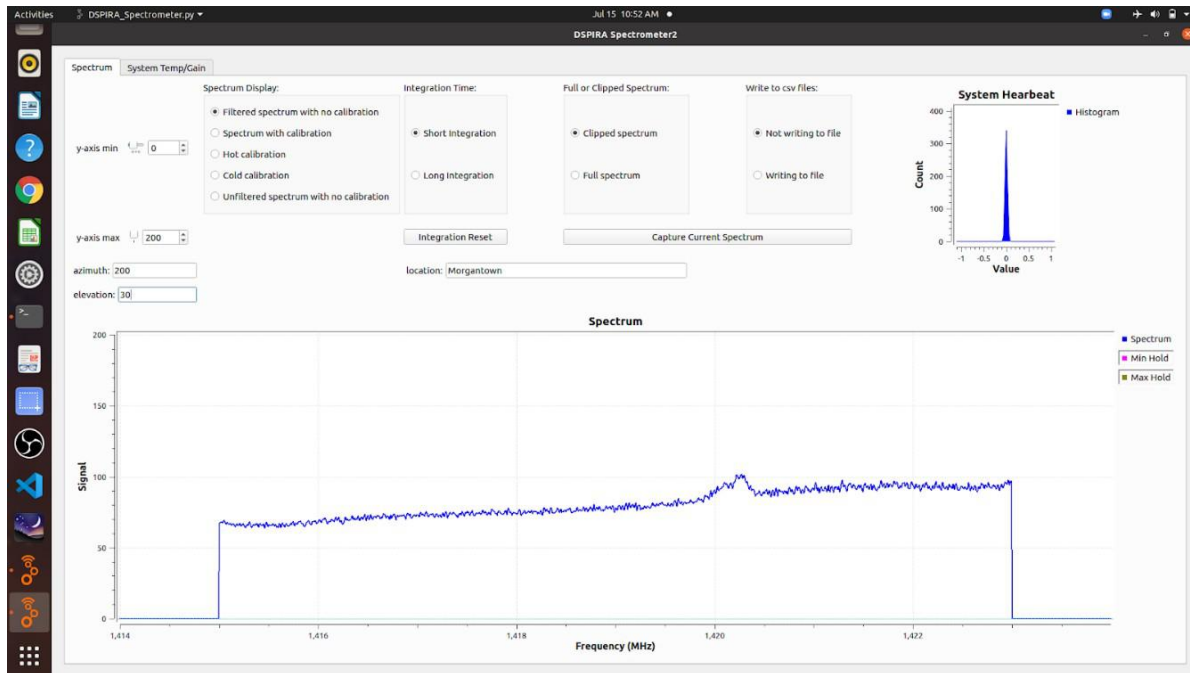


Figure 18. A typical observing display.

1. Adjust ymin and ymax to fit the spectrum on screen.
2. Enter the azimuth, elevation, and location. Press Enter after each entry so the program records it.
3. Use Short Integration while choosing a pointing. Switch to Long Integration when the telescope is settled.
4. Use Clipped Spectrum for routine viewing, or Full Spectrum when inspecting the band edges.

Save one spectrum or record continuously

Capture Current Spectrum saves the displayed spectrum as a CSV file. The prefix setting determines the destination folder.

Write to csv File saves a new file after each integration. A 10-second integration produces 60 files during a 10-minute run.

Continuous recording is useful for drift scans. Confirm the output folder and available disk space before leaving a run unattended.

Read saved data and calibrate the telescope

Saved spectra

CSV files open in spreadsheet software. They contain frequency in MHz and signal in two columns. Plot signal against frequency.

The filename includes the date, time, location, azimuth, and elevation, followed by `_spectrum`.

For example, `2020-07-15_11.06.14.0_Morgantown_200_30_spectrum` records a July 15, 2020 observation at 11:06:14. The pointing is 200° azimuth and 30° elevation.

Collect hot and cold references

Allow about 20 minutes for warm-up. Repeat calibration when conditions or the baseline change.

1. Point toward the ground. Select Hot Calibration, then Long Integration.
2. Adjust ymax if needed and wait for the spectrum to settle.
3. Switch to Cold Calibration before moving the telescope. This preserves the hot reference. Then select Short Integration.
4. Point at open sky. Find a patch without a hydrogen peak near 1420.4 MHz, then select Long Integration.
5. Wait for the display to settle. Switch to Spectrum with Calibration before moving the telescope to preserve the cold reference.
6. Inspect the baseline and hydrogen feature before collecting observations.

Hot Calibration and Cold Calibration continuously replace their respective references while selected. Spectrum with Calibration uses the saved references.

[Current calibration procedure and demonstration](#)

Check the calibrated baseline

Calibrated signal is expressed in kelvin (K). The model assumes a 10 K cold reference; it is not a universal sky temperature.

A drifting or negative baseline calls for checking the setup and repeating calibration. Changing the display scale does not correct calibration.

A constant offset does not shift peak positions. Frequency-dependent drift or interference can change the spectrum shape. Check stability before quantitative analysis.

Sources and related lessons

Construction text and illustrations were adapted from the DSPIRA mini horn construction guide. Operating instructions were adapted from Dr. John Makous, DSPIRA participant and physics teacher.

The original diagrams and interface screenshots are retained. Current software names, recording paths, and calibration guidance follow the maintained DSPIRA lessons.

Software checks confirm the reference-saving sequence. The full calibration procedure still needs a recorded hardware check.

Continue on the DSPIRA site

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[Install the current software](#)

[Spectrometer controls and demonstrations](#)

[Calibrate the horn telescope](#)

[Editable teaching resources](#)

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